

Algorithms

Previous (Two-Stage KAN) and Current (feature extraction+ KAN)

1. High-Level Summary

Aspect	Previous	Current
Detection Method	KAN with physics features	Feature extraction
Uses KAN for Detection?	Yes	No
Uses KAN for Localization?	Yes	Yes
Learning in Detection?	Learned (requires training)	Not learning by KAN
Number of KAN Models	2 (detection + localization)	1 (localization only)

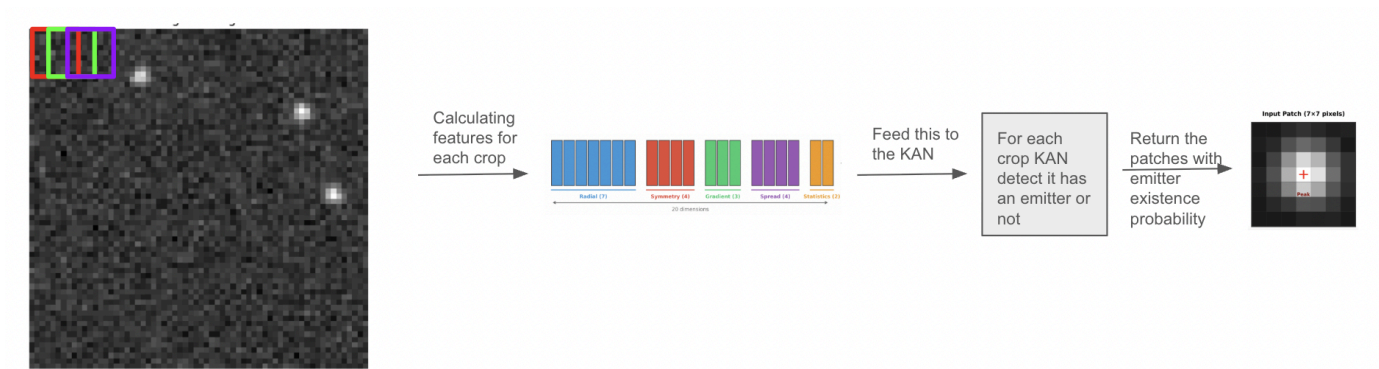
2. Detection Stage Comparison

Property	Previous	Current
Method	KAN neural network	Feature extraction
Input	20 physics-informed features	Raw pixel values
Features Used	Radial profile (7), Symmetry (4), Smoothness (3), Spread (4), Stats (2)	None (operates on pixels)
Architecture	$[20 \rightarrow 16 \rightarrow 8 \rightarrow 1]$	No neural network
Loss Function	Focal Loss	N/A
Blob Finding	Sliding window over all patches	DoG response + local maxima
Output	$p_{\text{emit}} \in [0, 1]$	Candidate list $[(x, y), \dots]$
Training Required?	Yes	No

3. Localization Stage Comparison

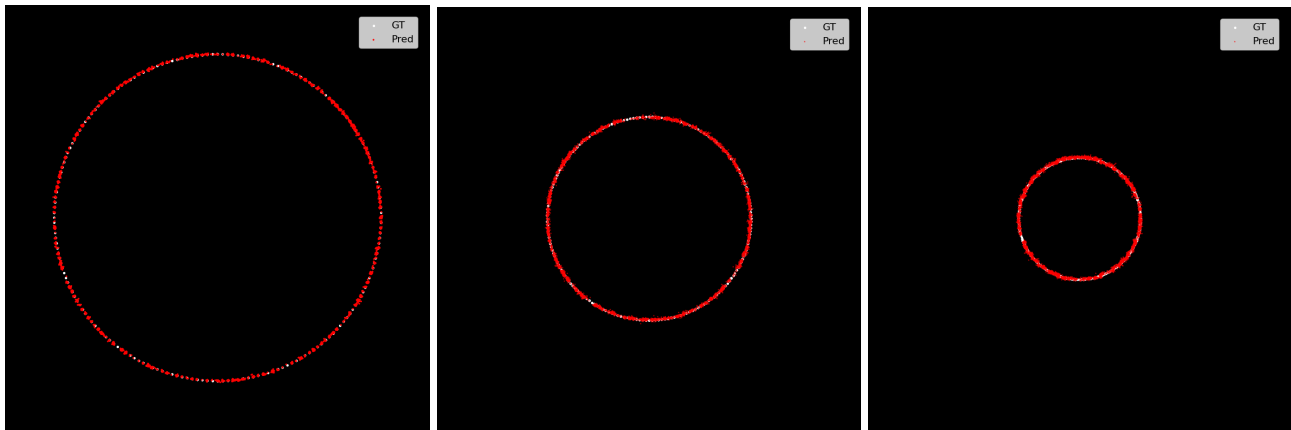
Property	Previous (V8.5)	Current (DoG + KAN)
Method	KAN neural network	KAN neural network
Crop Size	7×7 pixels	7×7 pixels
Input Size	49 raw pixels	32×32 = 1024 (resized)
Architecture	$[49 \rightarrow 64 \rightarrow 32 \rightarrow 2]$	$[1024 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 2]$
Grid Size	16	32
Loss Function	Wing Loss	Smooth L1 (Huber)
Output	(x, y) absolute position	(dx, dy) offset from anchor
Final Position	Direct output	anchor_center + offset × crop_nm
Training Required?	Yes	Yes

Previous method (sliding and stride) detection process:



dataset	UGIAF RMSMD	KAN RMSMD
LD 2D GUASSIAN	9.50	11.03
MD 2D GUASSIAN	21.55	12.41
HD 2D GUASSIAN	122.09	11.08

Reconstructed image:



Difference of gaussian method:

1. Gaussian Blur (Blur 1 - Wide Kernel): The original image is blurred with a large Gaussian kernel (large standard deviation,, removing fine details and noise.
2. Gaussian Blur (Blur 2 - Narrow Kernel): The original image is blurred again with a smaller Gaussian kernel (smaller (sigma), preserving more detail.Subtraction:
3. The heavily blurred image (Blur 1) is subtracted from the lightly blurred image (Blur 2).

- Result: The difference highlights areas where intensity changes rapidly (edges, blobs) but suppresses areas with high-frequency noise or very low-frequency backgrounds, resulting in a band-pass filtered image.

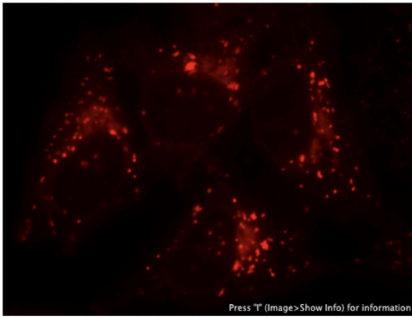


Fig. 161 Example input image

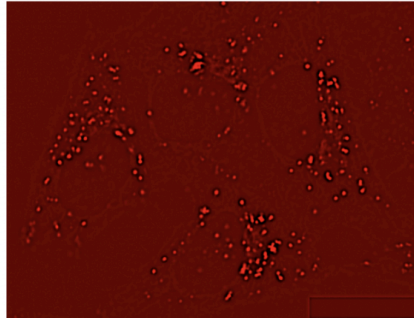


Fig. 162 Result of DoG filter

5.

STEP 1: Original Emitter (7×7 patch)

	100	102	105	110	105	102
	102	110	130	150	130	110
	105	130	180	220	180	130
	110	150	220	255	220	150
	105	130	180	220	180	130
	102	110	130	150	130	110
	100	102	105	110	105	102

← Peak = 255

STEP 2: Apply Gaussian blur ($\sigma=1.5$)

	108	112	118	122	118	112
	112	122	138	148	138	122
	118	138	168	188	168	138
	122	148	188	210	188	148

← Peak = 210 (reduced)

118	138	168	188	168	138	118
112	122	138	148	138	122	112
108	112	118	122	118	112	108

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STEP 3: Apply Gaussian blur ($\sigma=2.4$)

115	118	122	125	122	118	115
118	125	135	140	135	125	118
122	135	152	160	152	135	122
125	140	160	168	160	140	125
122	135	152	160	152	135	122
118	125	135	140	135	125	118
115	118	122	125	122	118	115

← Peak = 168 (more reduced)

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STEP 4: DoG = Blur($\sigma=1.5$) – Blur($\sigma=2.4$)

-7	-6	-4	-3	-4	-6	-7
-6	-3	3	8	3	-3	-6
-4	3	16	28	16	3	-4
-3	8	28	42	28	8	-3
-4	3	16	28	16	3	-4
-6	-3	3	8	3	-3	-6
-7	-6	-4	-3	-4	-6	-7

← Peak = 42 (STRONG!)

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Weights (Gaussian kernel):

0.01	0.02	0.03	0.03	0.03	0.02	0.01	
0.02	0.04	0.06	0.07	0.06	0.04	0.02	
0.03	0.06	0.09	0.10	0.09	0.06	0.03	
0.03	0.07	0.10	0.12	0.10	0.07	0.03	← Center: highest weight
0.03	0.06	0.09	0.10	0.09	0.06	0.03	
0.02	0.04	0.06	0.07	0.06	0.04	0.02	
0.01	0.02	0.03	0.03	0.03	0.02	0.01	

Example of noise in background:

Background with realistic noise + ONE hot pixel:

98	102	99	101	98	100	102	
101	99	103	100	102	99	98	
100	102	98	101	99	101	100	
99	100	101	[195]	100	98	102	← Hot pixel (195) in noisy background
102	98	100	99	102	100	99	
98	101	102	98	100	101	100	
100	99	98	101	99	102	98	

Background: fluctuates 98-103
noise pixel: 195 (sudden spike - only one pixel)

After blur $\sigma=1.5$:

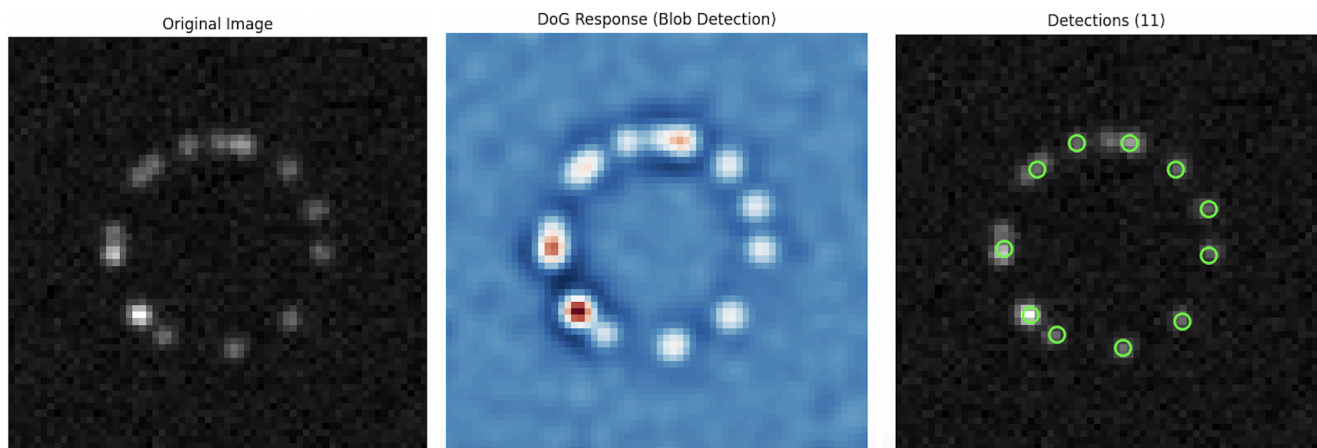
99	100	100	101	100	100	100	
100	100	101	103	101	100	99	
100	101	103	108	103	101	100	
100	101	106	[111]	106	101	100	← Was 195, now 111
100	100	102	106	102	100	100	
99	100	101	101	101	100	100	
100	100	99	100	100	100	99	

After blur $\sigma=2.4$:

99 100 100 100 100 100 100
100 100 101 102 101 100 100
100 101 103 105 103 101 100
100 101 104 [107] 104 101 100 ← Was 195, now only 107
100 101 102 104 102 101 100
100 100 101 101 101 100 100
100 100 100 100 100 100 100

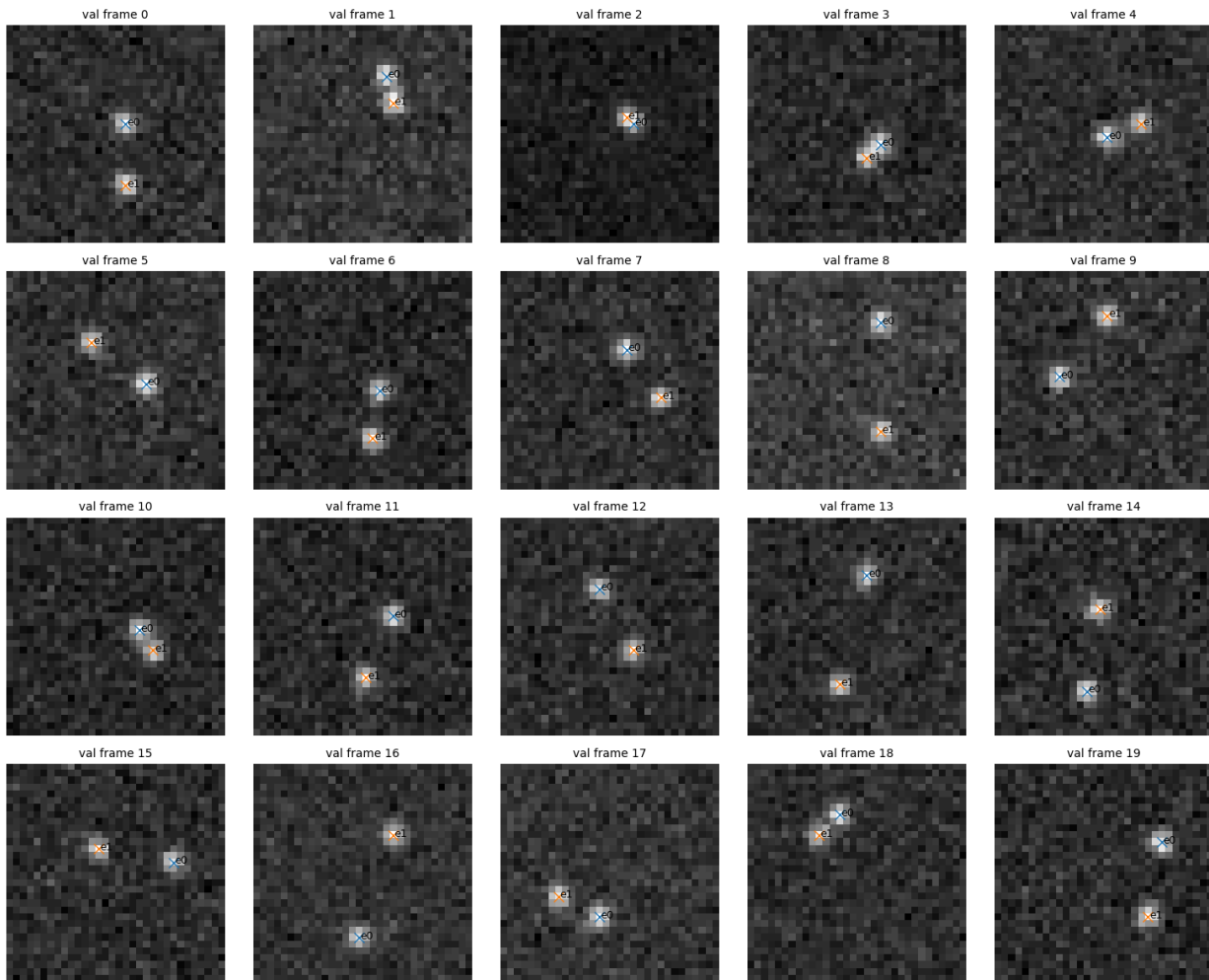
At hot pixel: $111 - 107 = 4$ ← WEAK response!

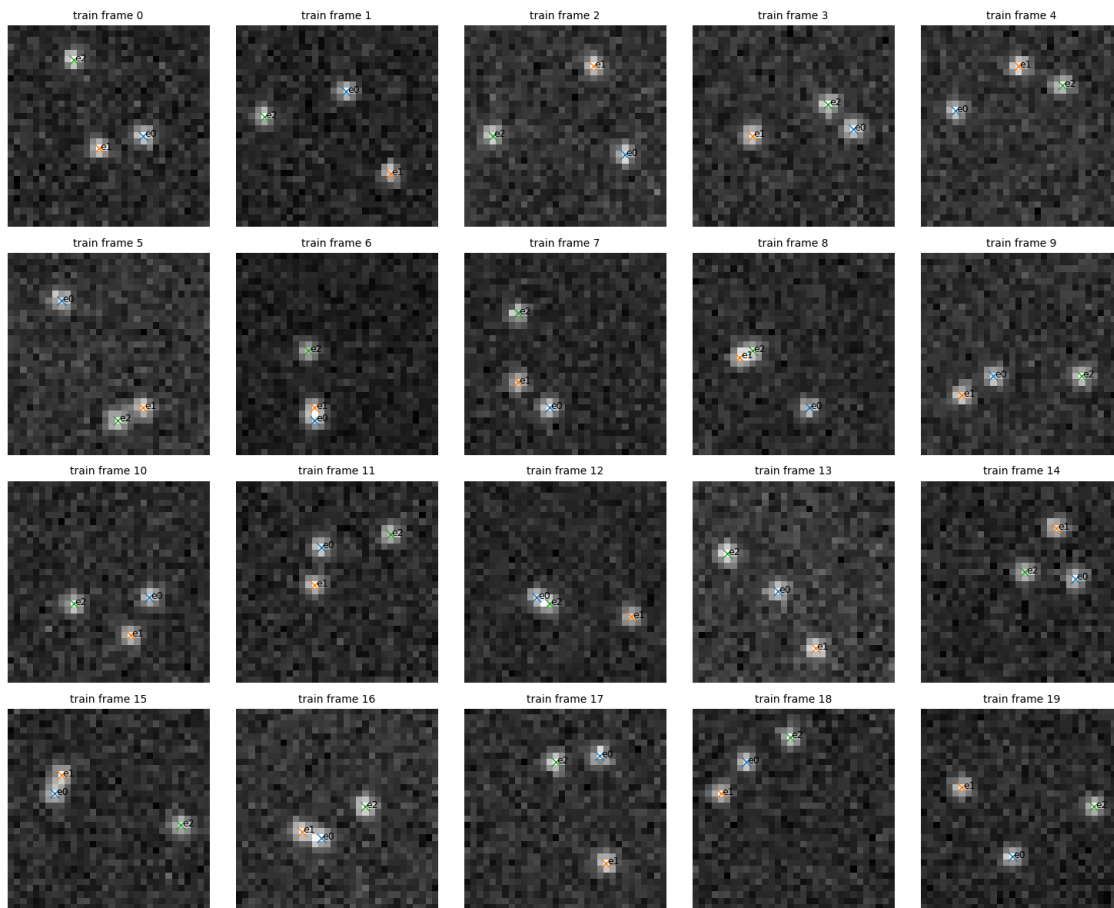
0 0 0 1 0 0 0
0 0 0 1 0 0 -1
0 0 0 3 0 0 0
0 0 2 [4] 2 0 0 ← Only 4! Too weak!
0 -1 0 2 0 -1 0
-1 0 0 0 0 0 0
0 0 -1 0 0 0 -1



Pure KAN Multi-Emitter Localization

2 emitters and 3 emitters (16*16) emitters in a circle of 5 pixels from the center





Feed through KAN $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 4$
 (No detection, no cropping, no post-processing)

f: pixel intensities $\rightarrow$ emitter positions Specifically:
 $[I_{00}, I_{01}, \dots, I_{15,15}] \rightarrow [X_1, Y_1, X_2, Y_2]$

256 inputs 4 outputs

The RMSMD is more than 3 pixels here.

What is the problem here? 2 emitters and no spatial structure

The solution here is using

Detection 256 pixels $\rightarrow$ 2 pixel locations

$\rightarrow$ Find peaks

KAN(localization) $\rightarrow$ sub-pixel-localization $\rightarrow$ small offset $\sim \pm 0.5$ pixel